An Algorithm for Aerosol Optical Properties Retrieval Over the Ocean Accelerated by a Neural Network From Single-View Multispectral Measurements of Intensity and Polarization

Zhe Ji, Zhengqiang Li, Cheng Fan, Cheng Chen, Zhenwei Qiu, Zhenhai Liu, Haoran Gu, Qian Yao, and Gerrit de Leeuw

***Abstract*—Monitoring aerosols over the oceans is critical for understanding Earth's climate and air quality. Although polarization can substantially reduce uncertainty in aerosol retrievals, current algorithms rely mainly on multiview polarimeters, and no dedicated algorithm is available for single-view polarimeters over the ocean. Here, we present the first ocean algorithm for a spaceborne single-view polarimeter, demonstrated with the particulate observing scanning polarimeter (POSP) onboard the GF-5(02) satellite. Our algorithm combines multispectral polarization with machine learning (ML)-accelerated radiative transfer (RT) calculation and seasonally clustered global aerosol models. Validation with Aerosol Robotic Network (AERONET) and maritime aerosol network (MAN) data demonstrates high accuracy, with RMSEs of 0.061, 0.479, and 0.037 for AOD_{550}, aerosol loading $(AE)_{670-870}$, and SSA_{550} using AERONET, and 0.030 and 0.259 for AOD_{550} and $AE_{670-870}$ using MAN, respectively. Comparison with retrievals from the generalized retrieval of atmosphere and surface properties (GRASPs) algorithm confirms that our algorithm performs comparably to GRASP products. These results underscore the necessity and feasibility of developing specialized aerosol retrieval algorithms for single-view polarimeters, and pave the way for global aerosol over the ocean monitoring.**

***Index Terms*—Aerosol retrieval algorithm, GF-5(02), ocean, particulate observing scanning polarimeter (POSP), single-view polarimetry.**

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I. Introduction

ATMOSPHERIC aerosols consist of droplets or particles of various forms and components suspended in the atmospheric fluid [1]. The formation mechanisms of aerosols are complex, and the sources of aerosols are diverse. Furthermore, aerosols have a short lifetime, typically ranging from a few hours to several days. These result in significant spatiotemporal heterogeneity in their size distributions and chemical compositions [2]. Aerosol particles are initially formed through nucleation, often catalyzed by atmospheric precursors. Subsequently, their size and chemical composition evolve continuously through processes such as condensation, coagulation, and cloud interactions [3]. Importantly, aerosols with different chemical compositions exhibit entirely distinct radiative properties [4]. Effective aerosol monitoring generally requires both high temporal and spatial resolution.

Ground-based observations offer high temporal resolution and highly accurate retrievals, but their spatial coverage is limited [5], [6]. Consequently, they are widely employed as reference data for validating satellite aerosol products. In contrast, satellite products are suitable for monitoring the aerosols on a large scale [7]. Nevertheless, their accuracy often remains insufficient for specific applications [8]. For radiative forcing assessment, Chen et al. [9] reported that the current Intergovernmental Panel on Climate Change (IPCC) reports underestimate the global aerosol radiative contribution. In addition, Lee et al. [10] emphasized that aerosols remain the largest source of uncertainty in radiative forcing estimates, primarily due to the insufficient accuracy of existing aerosol products. In addition, current satellite-based aerosol products

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are inadequate for atmospheric correction of high-resolution imagery [11] and for trace gas retrieval [12]. These limitations highlight the urgent need for more accurate aerosol retrieval algorithms to meet the growing demands of atmospheric environmental monitoring.

Satellite-based aerosol retrieval research dates back to the 1970s. Early algorithms exploited the low reflectance of the ocean surface in the red and near-infrared (NIR) bands to separate atmospheric and surface signals. Among the earliest sensors, the Advanced Very High Resolution Radiometer (AVHRR) has provided one of the longest continuous records, with data available since October 1978 [13]. However, AVHRR was not originally designed for aerosol monitoring; it offers only two VIS–NIR bands suitable for retrieval. Building on this limitation, Stowe et al. [14] proposed one of the pioneering aerosol retrieval algorithms using AVHRR measurements.

Another important sensor for retrieving aerosol properties over the ocean is the sea-viewing wide field-of-view sensor (SeaWiFS), an ocean color instrument launched aboard the SeaStar (OrbView-2) satellite on August 1, 1997. Gordon and Wang [15] proposed one of the first retrieval algorithms by simplifying the calculation of reflectance at the top of the atmosphere (TOA), treating the ocean surface as specular. Using the Cox–Munk model, they effectively estimated surface reflection, enabling separation of atmospheric and oceanic signals. Building on this, Sayer et al. [16] developed the SeaWiFS Ocean Aerosol Retrieval (SOAR) algorithm, which accounted for reflectance from whitecaps and adjusted the aerosol models accordingly. This algorithm was later optimized and adapted for the Visible Infrared Imaging Radiometer Suite (VIIRS) [17].

The successful launch of the Moderate Resolution Imaging Spectroradiometer (MODIS) marked a new era in satellite-based aerosol retrieval. MODIS was deployed aboard the Terra (1999) and Aqua (2002) satellites, offering 36 spectral bands and higher spatial resolution (0.25–1 km) than its predecessors. These advances substantially improved aerosol product quality. The early MODIS aerosol algorithms followed the AVHRR approach, but leveraged the broader spectral information [18]. Subsequently, Levy et al. [19] refined the algorithm by incorporating more realistic aerosol models and improving cloud screening and quality control. In recent years, with the continuous development of Chinese meteorological satellites, aerosol retrieval over the ocean has gradually approached the international level. Pei et al. [20] developed an ocean aerosol retrieval algorithm for the MERSI-II, filling the gap in ocean aerosol retrieval for the FY-3-D satellite.

Previous studies of aerosol retrieval primarily relied on the intensity measurements, but research by Mishchenko and Travis [21] has demonstrated that polarization provides distinct advantages over intensity alone in aerosol retrieval. Although several studies have incorporated polarization to retrieve aerosol properties over the ocean, these approaches have exclusively depended on multiview or multipixel observations [22]. To date, no dedicated algorithm has been developed for single-view polarimeters over the ocean. In this study, the particulate observing scanning polarimeter (POSP) onboard the GF-5(02) satellite, one of the few spaceborne single-view polarimeters, has been utilized. Building on our previous work that established an optimized retrieval framework for aerosol optical properties over land [23], [24], we extend and adapt this retrieval framework to enable accurate aerosol retrieval over the ocean.

To achieve higher retrieval efficiency, this study employs a neural network (NN) model to replace radiative transfer (RT) calculations. As Remer et al. [25] noted, the deep integration of artificial intelligence with aerosol remote sensing is becoming an inevitable trend, particularly through the substitution of physical process calculations with machine learning (ML) methods. Traditional optimization algorithms rely on a high-accuracy RT model, which is computationally expensive and unsuitable for real-time applications. By contrast, ML methods can approximate physical calculations with comparable accuracy while greatly reducing computational cost. For instance, Fan et al. [26] applied an NN model to replace RT calculation for water-leaving reflectance, Tao et al. [27] employed deep belief networks to substitute for RT calculation, and Di Noia et al. [28] used an NN model to provide first guesses for the Phillips–Tikhonov inversion algorithm. Collectively, these studies demonstrated that ML methods can substantially improve the efficiency of aerosol retrieval.

Section II describes the data used in this study. The methodology for aerosol retrieval over the ocean is presented in Section III. The new ocean aerosol model and its clustering process are presented in Section IV. The description of the algorithm is presented in Section V. The validation of the retrieval results is presented in Section VI. Section VII discusses the analysis of global aerosol distribution and the quantitative effect of adding polarization for aerosol retrieval. Finally, conclusions are provided in Section VIII.

II. Materials

A. POSP/GF-5(02) Data

The POSP is a single-view polarimeter specifically developed for aerosol retrieval. It is onboard the Chinese sun-synchronous satellite GF-5(02), launched on September 7, 2021. POSP employs a cross-track scanning mode, providing a spatial resolution of 6.4 km at nadir and a swath width of approximately 1850 km. As summarized in Table I, POSP measures the full Stokes vector from the ultraviolet to the shortwave infrared. The instrument is equipped with an onboard calibration system, achieving a radiometric calibration accuracy better than 5% and polarization calibration accuracy for the degree of linear polarization (DoLP) better than 0.005 [23].

B. AERONET Data

Aerosol Robotic Network (AERONET) is a global aerosol observation network aimed at obtaining benchmark aerosol data worldwide [6]. It utilizes the CE318 sun–sky radiometer to perform measurements of sun direct and sky scattered light at a frequency of every 15 min.

AERONET provides high-precision aerosol products in which the AOD product has a low uncertainty of approximately 0.01–0.02 [29]. Therefore, the AERONET product has been widely used for validating satellite aerosol products [30], [31]. AERONET offers three different quality levels of

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TABLE I
SPECTRAL SETTINGS FOR POSP/GF-5(02)

Band (nm)	Bandwidth (nm)	Polarization
380	20	Yes
410	20	Yes
443	20	Yes
490	20	Yes
670	20	Yes
865	40	Yes
1380	40	Yes
1610	60	Yes
2250	80	Yes

datasets. In this study, the strictly quality-assured Level 2.0 data are used for aerosol model clustering, while Level 1.5 data are used for validating satellite aerosol retrievals.

Maritime aerosol network (MAN) is a ship-borne observation network that serves as a supplement to AERONET sites. It measures multiband AOD aboard moving vessels using Microtops II sun photometers. The instruments undergo standard calibration procedures, and typically, the uncertainty of AOD does not exceed 0.02, which is slightly higher than that of AERONET [32].

On the one hand, the AERONET sites that are located within 20 km of the coastline or on islands were selected to validate the retrieval accuracy over coastal regions. On the other hand, the MAN measurements were used to evaluate the retrieval accuracy over open-ocean regions.

C. Ancillary Data

In this study, the ocean properties derived from other satellites were used to reduce the uncertainty of the calculation of sea surface reflectance during the retrieval. According to (3), the calculation of sea surface reflectance requires consideration of wind speed, salinity, and chlorophyll-a (Chl) concentration. In addition, the concentration of absorbing gases in the atmosphere is also used for gas absorption calibration. The wind speed and atmospheric gas concentration vary rapidly over time. Therefore, the hourly ERA5 reanalysis data were used. ERA5 was developed and maintained by European Centre for Medium-Range Weather Forecasts (ECMWF) [33]. In contrast, salinity and Chl concentration exhibit slower temporal variations. Thus, the monthly MODIS Chl concentration product [34] and the NASA Aquarius salinity product [35] were used. To further constrain the retrieval algorithm, the MODIS Collection 6.1 (C6.1) aerosol product MOD08_D3 was used to obtain the monthly global maximum and minimum AOD [36]. Moreover, to further validate the global distribution of POSP AOD, we also used the MODIS Collection 6.1 (C6.1) aerosol product MOD04 for comparison.

III. METHODOLOGY

A. Basic Principle

The satellite measurements can be expressed as a stock vector $[I, Q, U, V]^{\mathrm{T}}$, I refers to the intensity measurements, Q and U represent linear polarization, and V represents circular polarization. For measurements of the Earth's surface, V is always ignored [37], [38]. In this study, satellite measurements were normalized, including apparent reflectance (ρ) at the TOA and the DoLP

$$\rho = \frac{\pi I}{E_0 \mu_s} \tag{1}$$

$$\mathrm{DoLP} = \frac{\sqrt{Q^2 + U^2}}{I} \tag{2}$$

where E_0 refers to the extraterrestrial solar irradiance, and μ_s is the cosine of the sun's zenith angle.

B. Ocean Surface Reflectance

In the solar spectrum range, the ocean surface reflectance (ρ_{ocean}) for intensity is composed of three components [39], [40], specular reflection (ρ_{glint}) from the Cox–Munk model [41], reflection due to whitecaps (ρ_{foam}) [42], and water-leaving reflectance from the water body (ρ_{ul}) [26]

$$\rho_{\mathrm{ocean}} = W_s \rho_{\mathrm{glint}} + (1 - W_s)\rho_{\mathrm{foam}} + \rho_{\mathrm{ul}} \tag{3}$$

where W_s is the percent of area covered with whitecaps, Monahan and Muircheartaigh [43] used wind speed (W, units of m/s) to fit it as $W_s = 2.9510^{-6} W^{3.52}$, which is suitable when water temperatures are greater than 14 °C.

In addition, for the polarized reflectance of the ocean surface, the contributions from whitecap reflection and water-leaving reflectance are negligible. Therefore, only the contribution from specular reflection is considered.

1) Specular Reflection: Cox and Munk [41] constructed the sea surface glitter reflectance model as a function of wind speed through statistical methods. This reflectance depends on the positions of the sun and satellite, as well as the wind speed

$$\rho_{\mathrm{glint}} = \frac{\pi p\left(z'_x, z'_y\right) F\left(n, \theta_I\right)}{4 \cos\theta_s \cos\theta_v \cos^4\beta} \tag{4}$$

where θ_s is the zenith angle of the sun, θ_v is the zenith angle of view, and $p(z'_x, z'_y)$ is the probability distribution function of sea surface slopes in (S8), $F(n, \theta_I)$ is the Fresnel reflection matrix in (S1)–(S5), and β is the zenith angle of the inclined facet's normal direction, which can be derived using the sun zenith angle, view zenith angle, and the incident angle (θ_I)

$$\cos\beta = \frac{\cos\theta_s + \cos\theta_v}{2\cos\theta_I}. \tag{5}$$

2) Whitecaps Reflection: The reflectance of ocean surface whitecaps can be calculated by W_s and its corresponding effective reflectance (ρ_{ef}), which decreases with wavelength. Koepke [42] provided typical values for different spectral bands

$$\rho_{\mathrm{foam}} = W_s \rho_{\mathrm{ef}}(\lambda). \tag{6}$$

3) Water-Leaving Reflectance: The water-leaving reflectance (ρ_{ul}) refers to the radiation that leaves the water after being reflected by the water body and suspended materials. The optical properties of seawater are influenced by phytoplankton and their derivatives. Therefore, when

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assuming the sea surface to be Lambertian, the ρ_{ul} can be approximately estimated as [44] and [45]

$$\rho_{\mathrm{ul}} = \frac{1}{n^2} \frac{R_{\mathrm{w}}(\lambda)\, t_{\mathrm{u}}(\theta_s)\, t_{\mathrm{d}}(\theta_{\mathrm{v}})}{1 - aR_{\mathrm{w}}(\lambda)} \tag{7}$$

where n is the refractive index of sea water. a represents the albedo of water reflectance above the sea surface, with a typical value of 0.485 [46]. t_{u} and t_{d} are the transmittances of upwelling and downwelling radiation underwater, respectively. These transmittances can be obtained by integrating Fresnel's reflection coefficient. R_{w} is the ratio of upwelling to downwelling radiation beneath the water surface, which depends on the total absorption coefficient (A_λ) and the total backscattering coefficient (B_λ). Approximated values were provided by Morel and Prieur [45]

$$R_{\mathrm{w}}(\lambda) = 0.33 \frac{B_\lambda}{A_\lambda}. \tag{8}$$

C. Reflectance at the TOA

1) Intensity Reflectance at the TOA: The ocean surface is assumed to be a homogeneous Lambertian body, and the atmosphere is plane-parallel. Therefore, the intensity reflectance at the TOA can be expressed as [47] and [48]

$$\rho_{\mathrm{TOA}} = T_{\mathrm{g}} \left(\rho_0 + \frac{T^{\downarrow}(\mu_s)\, \rho_{ocean}\, T^{\uparrow}(\mu_{\mathrm{v}})}{1 - s\rho_{\mathrm{ocean}}} \right) \tag{9}$$

where T_{g} denotes the gas absorption transmittance, ρ_0 denotes the atmospheric path reflectance, and $T^{\downarrow}(\mu_s)$ and $T^{\uparrow}(\mu_{\mathrm{v}})$ are the atmospheric upward and downward transmittance, respectively. μ_s and μ_{v} are the cosines of the θ_s and θ_{v}, respectively. s represents the spherical albedo.

2) Polarization Reflectance at the TOA: The polarization reflectance at the TOA can be expressed as [49] and [50]:

$$\rho'_{\mathrm{TOA}} = T_{\mathrm{g}} \left(\rho'_0 + T'^{\downarrow}(\mu_s)\, T'^{\uparrow}(\mu_{\mathrm{v}})\, \rho'_{\mathrm{ocean}} \right) \tag{10}$$

where ρ'_0 denotes the polarized atmospheric path reflectance, and ρ'_{ocean} is the polarized reflectance of the ocean surface. Both $T'^{\downarrow}(\mu_s)$ and $T'^{\uparrow}(\mu_{\mathrm{v}})$ are related to the optical thickness of the atmosphere, which can be expressed as

$$T'^{\downarrow}(\mu_s) = \exp\left(-\left(\frac{\psi\tau_\lambda^m + \varsigma\tau_\lambda^a}{\mu_s} \right) \right) \tag{11}$$

$$T'^{\uparrow}(\mu_{\mathrm{v}}) = \exp\left(-\left(\frac{\psi\tau_\lambda^m + \varsigma\tau_\lambda^a}{\mu_{\mathrm{v}}} \right) \right) \tag{12}$$

where τ_λ^a and τ_λ^m represent the aerosol and molecular optical thickness. The coefficients ψ and ς are used to reduce the impact of diffuse interactions between the surface and the atmosphere [50]. The definitions and values of the coefficients ψ and ς can be found in the supplementary materials.

We used the RT code 6SV2.1 to calculate the atmospheric parameters in (9) and (10) under different observation geometries. The output from 6SV2.1 includes path reflectance, atmospheric transmission, and the spherical albedo. These parameters are stored in look-up tables (LUTs), which can be interpolated for a given observation geometry to compute the corresponding apparent reflectance. The details on the construction of the LUTs can be found in the supplementary materials.

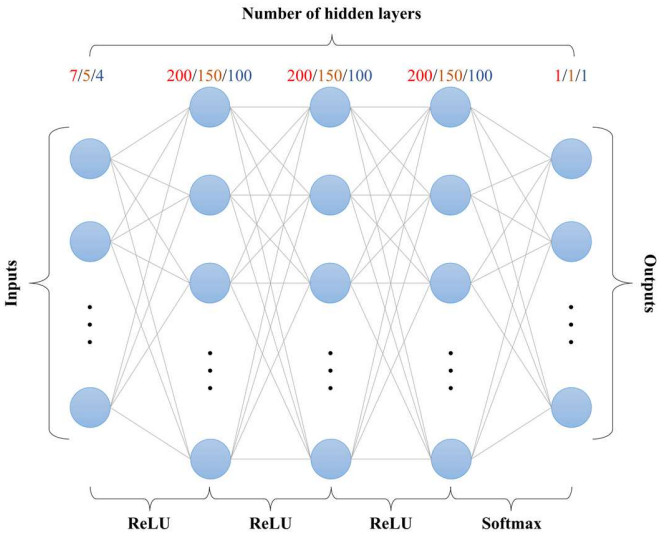


Fig. 1. Structure of the NN in this study. The red, brown, and blue numbers indicate the intensity reflectance at the TOA, polarization reflectance at the TOA, and atmosphere gas absorption transmittance, respectively.

D. NN Model

The NN model used in this study was implemented using the Python library scikit-learn. After hyperparameter optimization, we obtained a five-layer NN model, as shown in Fig. 1, consisting of one input layer, three hidden layers, and one output layer. The number of neurons in the three hidden layers was set to 200, 150, and 100, respectively, depending on the output parameters. For the model that outputs intensity reflectance at the TOA, the inputs include sea surface wind, sea surface salinity, Chl concentration, solar zenith angle, view zenith angle, relative azimuth angle, and aerosol optical depth. For the model that outputs polarized reflectance at the TOA, the inputs include sea surface wind, solar zenith angle, view zenith angle, relative azimuth angle, and aerosol optical depth. For the model that outputs atmosphere gas absorption transmittance, the inputs include gas content, solar zenith angle, view zenith angle, and relative azimuth angle.

Before training, input features were normalized, and an NN model was trained separately for each aerosol model. During training, the learning rate was set to 0.0001, and the Adam optimizer was used. A Sigmoid function was applied after the output layer, while ReLU functions were used as activation functions between the other hidden layers. The loss function was defined as the mean squared error (mse), and early stopping was employed to prevent overfitting. For validation, observation geometries uniformly distributed across the parameter space were randomly selected to form the test dataset. As shown in Figs. S1 and S2, the validation results demonstrate that the R^2 consistently exceeds 0.95 for all parameters.

E. Inversion Algorithm

Unlike traditional optimization algorithms [37], [51], we used the Gauss–Newton iteration method without incorporating regularization, but instead constrained the optimization

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results through boundary conditions, as follows:

$$\hat{x} = \left[\| \mathbf{S}_{\mathrm{y}}^{-\frac{1}{2}} (\mathbf{y} - \mathbf{F}(\mathbf{x})) \|^2 \right] \tag{13}$$

where $\hat{x}$ refers to the optimal estimate of the state vector $\mathbf{x}$, $\mathbf{S}_y$ is the error matrix from the instrument, and $\mathbf{y}$ and $\mathbf{F}(\mathbf{x})$ refer to the measurements and the simulations, respectively. The iterative solution is given by

$$\tilde{x}_{n+1} = \tilde{x}_n + \Lambda \left(\tilde{K}^{\mathrm{T}} \tilde{K} \right)^{-1} \tilde{K}^{\mathrm{T}} \tilde{y} \tag{14}$$

where $\tilde{y} = \mathbf{S}_{\mathrm{y}}^{-\frac{1}{2}}(\mathbf{y} - \mathbf{F}(\tilde{x}_n)), \tilde{K} = \mathbf{S}_{y}^{-\frac{1}{2}}\mathbf{K}, \Lambda$ refers to the filter factor, $\mathbf{K}$ refers to the Jacobian matrix, and n is the iteration step. The retrieval requires an initial guess; therefore, we use random values between 0 and 0.5 as the initial input to drive the iteration beginning. For each iteration, the end of the iteration is determined via a goodness-of-fit assessment by comparing χ^2 as given by

$$\chi^2 = \frac{1}{N} \sum_{i=1}^{N} \left(\frac{\boldsymbol{y}_i - \boldsymbol{F}_i}{e_i} \right)^2 \tag{15}$$

where N is the number of measurements, including the ρ_λ and DoLP_λ, and e_i is the measurement uncertainties from the instrument. When the $\chi^2_{\min} < 5$, the iteration is finished.

In addition, boundary constraints are defined by empirical knowledge and instrument errors. On the one hand, to reduce the impact of instrument errors on the retrieval, the calibration accuracy is used for the boundary conditions: for reflectance, $\mathrm{abs}(\rho_i - \hat{\rho}_i)/\rho_i < e_{\rho_i}$; for DoLP, $\mathrm{abs}(\mathrm{DoLP}_i - \widehat{DoLP}_i) < e_{DoLP_i}$. On the other hand, the retrieved AOD has seasonal characteristics [52], [53], [54]. Therefore, the AOD to be inverted should fall within a certain statistical range. The MOD08_M3 (from 2019 to 2024) has been used to determine the possible AOD ranges.

F. Aerosol Properties

In this study, the two log-normal modes have been used for aerosol volume size distribution (V) [55]

$$V(r) = \sum_{i=1}^{2} \frac{1}{\sqrt{2\pi}\sigma_{\mathrm{g},i} r} \exp\left(-\frac{1}{2} \left(\frac{\ln r - \ln r_{\mathrm{g},i}}{\sigma_{\mathrm{g},i}} \right)^2 \right) \tag{16}$$

where r_{g} refers to the mean radius, and σ_{g} represents the standard deviation. The two log-normal modes include one fine mode and one coarse mode.

Furthermore, the complex refractive index (m) of the aerosol mode can be expressed as

$$m = m_r + m_i i \tag{17}$$

where m_r and m_i represent the real and imaginary parts, respectively. It should be noted that, in this study, the complex refractive index of the fine mode and coarse mode is the same.

IV. Ocean Aerosol Models

A. Cluster Results of Aeronet Data

Fig. 2 presents the AERONET sites employed for aerosol model clustering, comprising 381 coastal sites and 73 island sites. To guarantee sufficient data for clustering, Level 2.0 AERONET data spanning 23 years (2002–2024) were utilized.

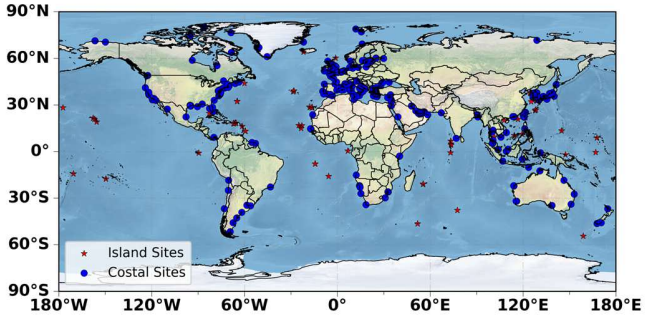


Fig. 2. Global distribution of AERONET sites used for aerosol model clustering. Blue circles represent coastal sites, while red pentagrams indicate island sites.

To reduce the uncertainty in the clustering results, only observations with AOD at 550 nm exceeding 0.4 were used [56].

The classification of coastal and island sites was manually determined through visual inspection. Specifically, sites located on isolated islands or at a considerable distance from the mainland were classified as island sites. In contrast, those within a 20-km coastal buffer zone were classified as coastal sites.

Due to seasonal variations in atmospheric conditions, this study performed clustering on AERONET data separately for each season. The k-means method, which has been widely used for clustering the empirical aerosol models [57], [58], was employed in this study. Specifically, clustering was conducted based on the volume, mean radius, standard deviation, and complex refractive indices at various wavelengths from AERONET. After testing and balancing clustering accuracy and model applicability, the optimal number of clusters was determined to be one for island sites and four for coastal sites.

Fig. 3 shows the percentage of each clustered aerosol model at each AERONET site per season. Due to observational limitations, the number of available AERONET sites during DJF is significantly lower than in other seasons. Tables S2–S6 present the optical properties of the clustered aerosol models. It is important to note that the same aerosol type number in different seasons does not represent similar aerosol characteristics. The complex refractive indices of the clustered aerosol models generally fall within the range of 1.33–1.55. The mean radius of the fine mode is less than 0.2 μm, while that of the coarse mode is close to 2 μm. These results are consistent with previous studies on marine aerosols [16], [17].

This study further analyzed the dominant aerosol type at each AERONET site per season, as shown in Fig. 4. It can be observed that northern Africa and eastern Asia generally exhibit similar aerosol characteristics across different seasons. In contrast, Europe appears to have more similar aerosol properties during December–January–February (DJF) and March–April–May (MAM) than during June–July–August (JJA) and September–October–November (SON). In addition, the east and west coasts of both Americas and Africa tend to exhibit distinct aerosol characteristics.

B. Physical Properties of the Aerosol Models

Based on the clustered optical properties of the aerosol models, we calculated the aerosol size distribution and phase

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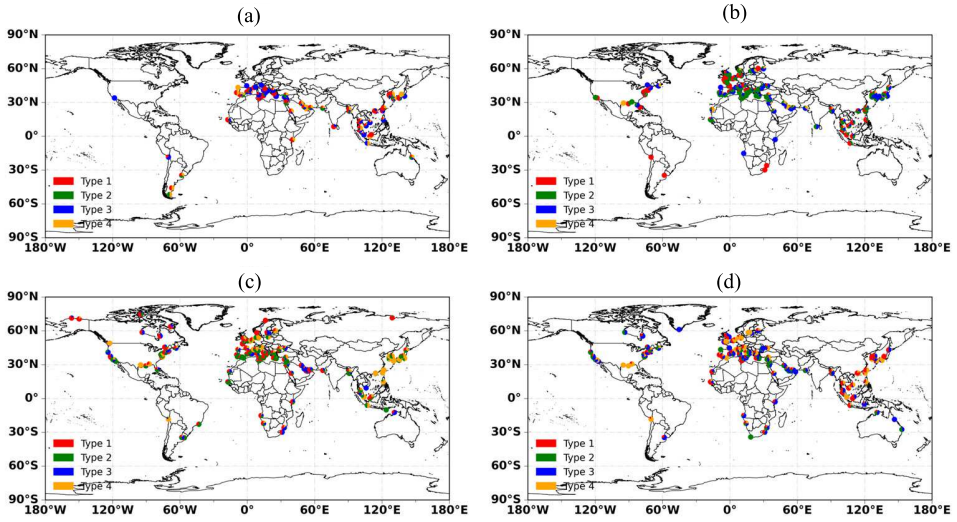


Fig. 3. Percentage of clustered aerosol model type at each AERONET site for (a) DJF, (b) MAM, (c) JJA, and (d) SON. Colors represent the different aerosol model types.

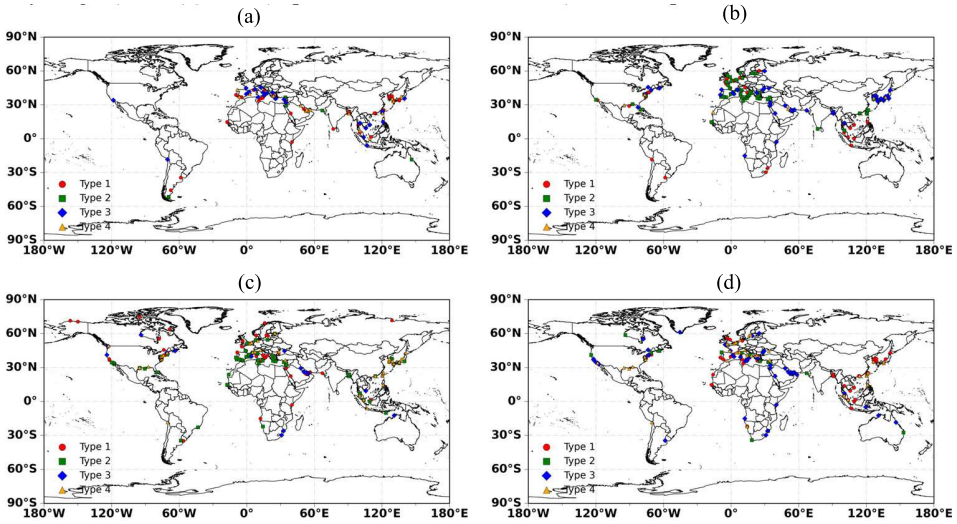


Fig. 4. Same as Fig. 3, but for the final aerosol model type at each AERONET site. (a) DJF. (b) MAM. (c) JJA. (d) SON.

function for different aerosol models, as shown in Figs. 5 and 6, respectively.

The coarse-mode volume during DJF is lower than that in other seasons, while the fine-mode volume is similar to that in other seasons. This phenomenon may be attributed to the lower temperatures in winter, which reduce the volatility of sea salt aerosols. In addition, the lower humidity during winter also limits the release of sea salt [59].

The clustered aerosol type from AERONET sites over islands is dominated by the coarse-mode aerosol in all seasons, and the complex refractive index is around 1.45. This suggests that the aerosol type for islands is mainly composed of sea salt. Affected by relative humidity, the refractive index of sea salt decreases from 1.5 in dry conditions to about 1.45 [60]. Therefore, we consider the aerosol type for islands to be reasonable. In contrast, the aerosol type for coastal oceans is more compositionally complex, due to its proximity to aerosol sources on land. Both Tables S3 and S5 show that during DJF, there are strongly absorbing aerosol types, with the imaginary part of the complex refractive index exceeding 0.01. During DJF, most of the available AERONET sites are located in the Northern Hemisphere, which is in winter at that time. So the presence of strongly absorbing aerosols may be attributed to anthropogenic emissions [61], [62].

Mie scattering theory was used to calculate the scattering phase functions of different aerosol types. Fig. 6 presents the results for DJF, while Figs. S3–S5 correspond to the other seasons. For the scattering phase function of fine-mode aerosols, significant differences are observed in the polarized scattering phase function (P_{12}) among different aerosol types. In contrast, for coarse-mode aerosols, the differences in P_{12} are mainly found at scattering angles less than 60° and greater than 150°. The scalar scattering phase function (P_{11}) of both coarse- and fine-mode aerosols shows similar patterns. When the scattering angle exceeds 120°, noticeable differences in P_{11} appear among the various aerosol types.

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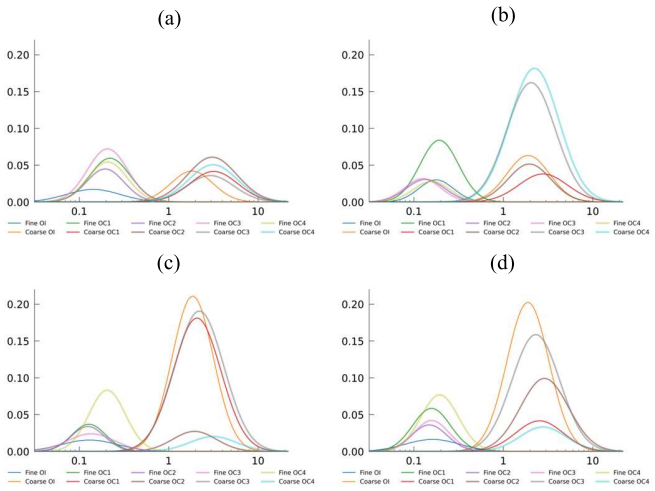


Fig. 5. Aerosol size distribution for different aerosol model types. OI represents the aerosol model over an ocean island, and OC represents the aerosol model over the coastal ocean. (a) DJF. (b) MAM. (c) JJA. (d) SON.

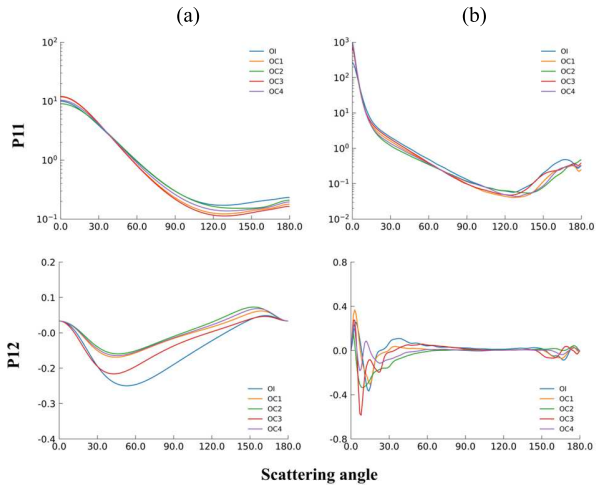


Fig. 6. Scattering phase function for different aerosol model types in DJF. OI represents the aerosol model over an ocean island, and OC represents the aerosol model over the coastal ocean. (a) Fine mode. (b) Coarse mode.

TABLE II
SUMMARY OF SCREENING THRESHOLDS

Items	Purposes
$\rho_{490} > 0.4$	Cloud
$\sigma_{443} > 0.0025$	Cloud
$\rho_{1380} > 0.03$	Cloud
$\rho_{glint} > 0.005$	Sun glint
Chl > 3 mg·m^{-3}	Turbid
$abs(\hat{\rho}_{670} - \rho_{670}) > 0.02$ or $abs(\hat{\rho}_{865} - \rho_{865}) > 0.02$	Shallow

V. SYNOPSIS OF THE RETRIEVAL ALGORITHM

A. Procedure of Retrieval

In previous studies, we proposed a high-accuracy aerosol retrieval framework over land [23], [24]. Building upon this framework, we optimized it based on the characteristics of the ocean aerosol. Most of the part for retrieval was retained. But the cost function is replaced by (13) to achieve higher retrieval accuracy. In addition, the aerosol model was also replaced with the clustered ocean aerosol model mentioned above. Fig. 7 presents the flowchart of the algorithm. To ensure the reliability of the retrieval, pixels that are likely to have high retrieval uncertainty (i.e., affected by cloud contamination, sun glint, or reflection from the ocean bottom) are excluded before retrieval. The thresholds used for excluding such pixels are listed in Table II. ρ and $\hat{\rho}$ denote the measured and simulated reflectance at the TOA, respectively, with the subscript indicating the corresponding band.

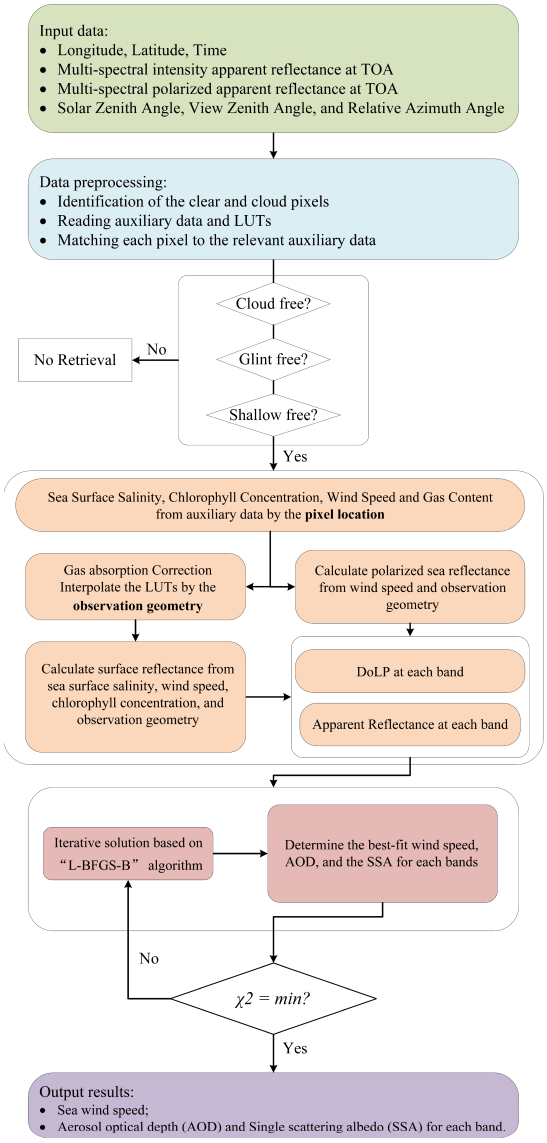


Fig. 7. Flowchart of the POSP aerosol retrieval algorithm over the ocean.

For the pixels to be retrieved, gas absorption correction is first applied to the satellite measurements. The detailed correction methods are described in Ji et al. [23]. The scalar and polarized reflectances at TOA are simulated using (9) and (10), respectively. Then, the DoLP is calculated using (2). Finally, the optimal retrieval results are iteratively obtained using the optimization method introduced in Section III-E.

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It should be noted that during the retrieval process, each aerosol model yields a cost function value. Therefore, the retrieval result corresponding to the minimum cost function is selected as the final output, and the associated aerosol model is considered the most probable aerosol type.

B. Cloud Mask

Cloud detection algorithms originally developed for MODIS have been widely adapted for other satellites [63]. In this study, the cloud detection strategies of MODIS were applied to POSP over the ocean, specifically, employing the apparent reflectance threshold and spatial variability methods [64].

On the one hand, the apparent reflectance threshold method is effective in identifying thick clouds with high brightness or significant water vapor content. Our tests showed that pixels can be effectively identified as cloudy when the reflectance at 490 and 1380 nm exceeds 0.4 and 0.03, respectively.

On the other hand, the apparent reflectance spatial variation detection method is suitable for detecting cloud edges, cloud shadows, thin clouds, and scattered clouds. Unlike cloud detection algorithms over land [23], this study uses only the spatial variation of the 443-nm band to detect clouds. A pixel is considered contaminated by clouds when the spatial variation exceeds 0.0025.

C. Sun Glint Mask

The sun glint reflectance can be estimated using the wind speed by the Cox–Munk model in (4). In this study, the same threshold as in Sayer et al. [17] is applied: The pixels with glint reflectance over 0.005 are excluded from the retrieval.

D. Turbid/Shallow Water Mask

Shallow seawater can lead to ocean surface reflectance containing contributions from the ocean bottom, introducing substantial uncertainty in surface reflectance estimation. In this study, we employed the algorithm of Li et al. [65] to identify shallow waters, which has been widely applied for MODIS, SeaWiFS, and VIIRS [16], [17].

When the reflectance at 670 or 865 nm does not conform to a power–law fit between the reflectance and wavelength in the blue (490 nm) and shortwave infrared bands (1610 and 2250 nm), the pixel will be considered shallow water. Our tests show that if the difference between the observed and expected reflectance at 670 and 865 nm exceeds 0.02, the pixel can be effectively identified as shallow water.

Moreover, when the water is turbid, the estimation of water-leaving reflectance in (7) may cause large errors. To reduce the impact of this condition on the retrieval, the same threshold as in Sayer et al. [17] is applied: pixels with Chl concentration over 3 mg $\cdot$ m^{-3} are excluded from the retrieval.

VI. Validation

A. Matchup Methodology

In this study, the accuracy requirements of the Global Climate Observing System (GCOS) were used to validate the aerosol retrievals. The definition of GCOS requirements can be found in Hasekamp et al. [66]. For AOD, the error should be smaller than 0.04% or 10% (whichever is greater). For SSA, the error should be smaller than 0.03. The GCOS requirements are not formulated for aerosol loading (AE). The other definition of statistical metrics can be found in (S17)–(S20).

Fig. 8 shows the POSP/AERONET and POSP/MAN matchups used for validation, with the color of the circles indicating the annual mean AOD_{550} observed at each location. The AERONET sites and the MAN cruises were used to validate the retrieval accuracy over coastal regions or open oceans, respectively. In general, coastal areas are more affected by human activities than open-ocean regions, resulting in higher Chl concentrations, which makes the aerosol retrieval more challenging. Thus, the validation accuracy between POSP and AERONET is always lower than the results between POSP and MAN [70]. Our validation results also confirm this.

To evaluate the accuracy of aerosol parameters retrieved, 12 months of data were used, obtained over the ocean between January 2022 and December 2022. Although the validation of aerosol retrieval over land generally uses the average of satellite retrievals within a 20-km radius around the AERONET site [31], [67], [68], [69], this approach may not be feasible for validating aerosol retrievals over coastal regions due to the presence of land. Therefore, the validation range was extended to 50 km [70]. The validation range for MAN cruises is still 20 km.

In addition, since AE changes more rapidly than its physical properties. Thus, for AOD and AE, AERONET data within ± 1 h of the satellite overpass were averaged; whereas for SSA (physical property), AERONET data within ± 3 h of the overpass were averaged. To minimize uncertainties introduced by averaging and to ensure statistical significance. The match was considered successful only when at least two AERONET observations were available within the temporal window and more than three valid satellite retrievals were present within the spatial window [67].

B. AERONET Coastal and Island Sites

Fig. 9 presents the validation results for aerosol parameters (AOD_{550}, $AE_{670-865}$, and SSA_{550}) retrieved from POSP over the AERONET coastal and islands sites. The biases for all parameters are close to zero, indicating that the retrievals are nearly unbiased.

For AOD_{550}, the RMSE is 0.061 and the MAE is 0.04, with 66.7% of the retrievals meeting the GCOS accuracy requirements. Notably, for AOD <0.2, the fraction within GCOS is higher than the overall validation.

AE and SSA validations were performed for $AOD_{550} > 0.1$. Although AE and SSA accuracies are lower under low AE, the AOD threshold of 0.1 was adopted to ensure a sufficient number of retrievals for validation.

For $AE_{670-865}$, AE is calculated using the AOD retrieved at 670 and 870 nm: $AE_{670-870} = -\frac{\ln[AOD_{670}-AOD_{870}]}{\ln[670/870]}$. Overall, the uncertainty of the AE retrieved from POSP is low, with an RMSE of 0.479 and a MAE of 0.379.

The retrievals are concentrated when AE from AERONET is lower than 0.75. This indicates that the retrievals are generally composed of coarse-dominated aerosol particles.

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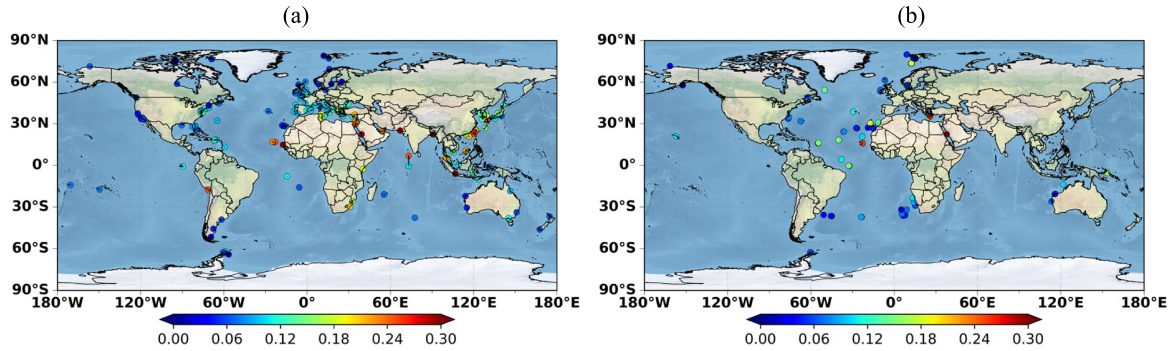


Fig. 8. Locations of POSP/AERONET and POSP/MAN matchups. Points where the yearly averaged AOD at 550 nm. (a) AOD_{550} at AERONET. (b) AOD_{550} at MAN.

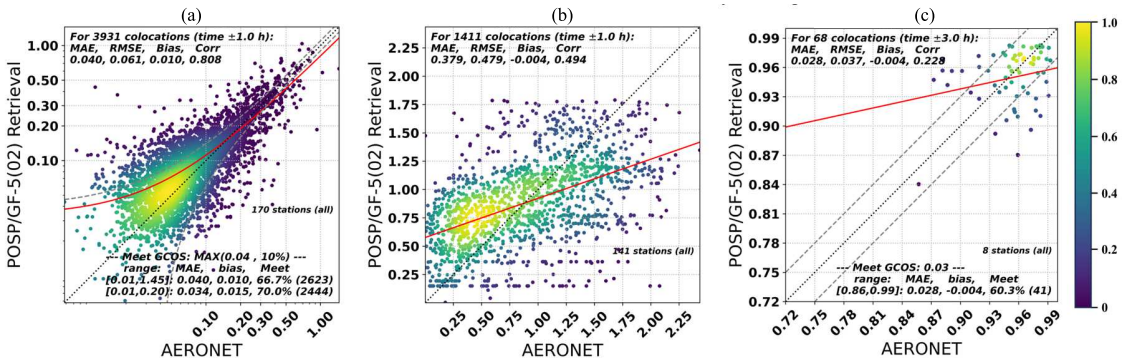


Fig. 9. Validation of POSP retrievals results over the ocean using AERONET as reference (a) for AOD at 550 nm, (b) for AE between 670 and 870 nm, and (c) for SSA at 870 nm. The black dashed line is the 1:1 reference line. The red solid line is the regression line. The gray dashed line represents the GCOS uncertainty envelopes. The color means the kernel density for scatter plots.

For SSA_{550}, the AEROENT SSA is calculated based on the AOD and aerosol absorption optical depth (AAOD) at 550 nm: $SSA_{550} = 1 - \frac{AAOD_{550}}{AOD_{550}}$. The validation with AERONET shows high accuracy with an RMSE of 0.037% and 60.3% of the retrievals meet the accuracy requirements set by GCOS.

C. MAN Cruises

The MAN does not provide SSA results. Therefore, only AOD and AE can be validated over the open ocean. It should be noted that, due to the influence of sun glint and the limited observation frequency of shipborne measurements, the number of matched samples between POSP and MAN is significantly lower than that between POSP and AERONET. Furthermore, in this study, the negative AE values were used for validation, which resulted in an inconsistency in the number of matchups for AOD and AE. Fig. 10 shows the validation results for aerosol parameters retrieved from POSP (AOD_{550} and $AE_{670-865}$) over the MAN cruises. The results demonstrate that the proposed algorithm in this study achieves high accuracy over open-ocean areas. For AOD_{550}, the RMSE is 0.03, and the bias is close to zero. Furthermore, 81.2% of the retrievals were within the GCOS. For $AE_{670-865}$, the RMSE is 0.259 and the MAE is 0.223.

Validation results from AERONET and MAN indicate that the retrieval errors mainly arise from two aspects. On the one hand, the accuracy of estimation for ocean surface reflectance affects the retrieval accuracy. For MAN, the accuracy is notably higher than for AERONET, primarily because the open-ocean surface is more homogeneous, less influenced

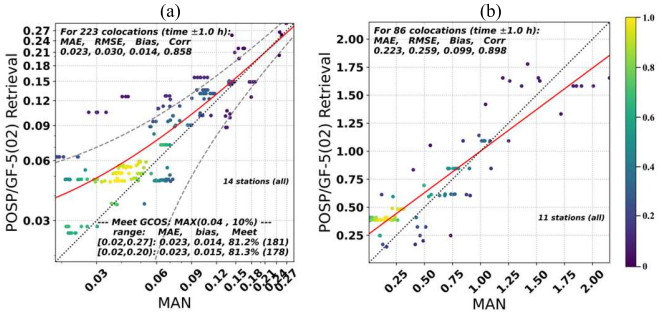


Fig. 10. Same as Fig. 9, but using MAN as a reference. (a) AOD_{550}. (b) $AE_{670-865}$.

by human activities, and characterized by lower Chl concentrations. In addition, the AERONET sites over coastal areas may suffer from undetected shallow water, which further reduces the accuracy of water-leaving reflectance estimation. On the other hand, the use of empirical aerosol models also contributes to retrieval uncertainty. Coastal regions are closer to the continental aerosol sources, resulting in more complex aerosol compositions compared to open oceans.

VII. Discussion

A. Global Analysis

Fig. 11 shows the global distribution of aerosol optical properties. AE over oceans is generally lower than over land. Considering the considerable computational cost of the optimal retrieval, we performed the retrieval only for the first

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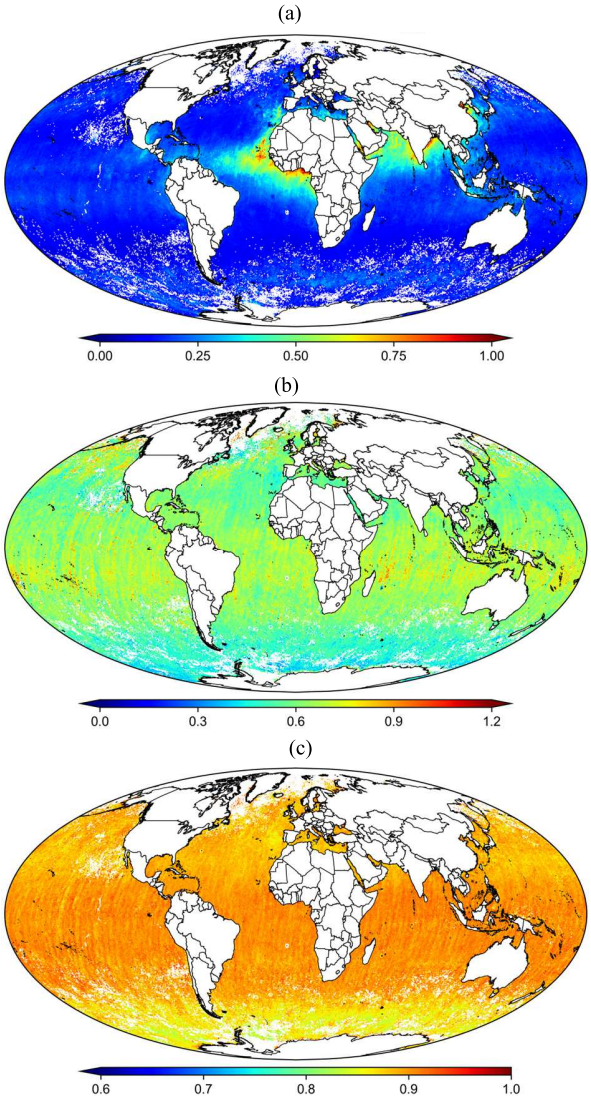


Fig. 11. Annual global spatial distributions of (a) AOD at 550 nm, (b) AE between 670 and 865 nm, and (c) SSA at 550 nm, based on POSP/GF-5(02) data from January 1, 2022 to December 31, 2022.

ten days of each month. Fig. 11(a) shows that the high AE over the ocean is mainly concentrated near coastal areas. The AOD is high over southern India and eastern China, particularly in the Bohai Bay. This is because aerosols from combustion or industrial processes are primarily located on land, and most aerosol particles can only be transported to nearby oceanic regions. The long-range aerosol transport can also be clearly observed over the Atlantic Ocean. The dust aerosols from North Africa are transported to the ocean. Fig. 11(b) shows that the global $AE_{670-865}$ over oceans is mainly concentrated around 0.6. This result is consistent with the validation results shown in Fig. 9. Similar to the work of Chen et al. [71], most of the SSA_{550} over the ocean exceeds 0.9, which is attributed to the marine aerosols composed of sea salt. It is worth noting that Chen et al. [71] did not provide retrieval results for regions near Antarctica. For our retrievals, the $AE_{670-865}$ and SSA_{550} over the ocean near Antarctica are lower than in other regions. On the one hand, the assumption of *a* plane-parallel atmosphere becomes less valid under large sun zenith angles. On the other hand, the AE over the Antarctic region is very low, providing limited information for retrieval.

TABLE III
COMPARISON OF AOD_{550} BETWEEN POSP AND MODIS

Parameter	RMSE	MAE	GCOS (%)
POSP	0.061	0.040	66.7%
MODIS	0.088	0.059	49.2%

Both factors increase the retrieval difficulty in these regions. We will address this issue in future studies.

To assess the accuracy of the AOD spatial distribution retrieved from POSP. Fig. 12 shows the comparison with MODIS/TERRA products, which is close to the POSP/GF-5(02) overpass time. The comparison indicates that POSP and MODIS AOD are generally consistent, with the annual global mean difference within 0.2. Exceptions are observed over the western North African ocean region, where POSP retrievals are notably higher than MODIS. Globally, POSP AOD tends to exceed MODIS AOD in most regions. This is expected, as the spatial resolution of POSP (6.4 km) is considerably lower than that of MODIS (0.25–1.0 km), increasing the likelihood that individual POSP pixels contain undetected clouds. Previous studies have shown that the presence of broken clouds can lead to an overestimation of retrieved AOD [14].

B. Comparison of AOD Between POSP and MODIS

The deep blue (DB) algorithm does not provide aerosol products for MODIS over the ocean. So we used the ocean aerosol products from the dark target (DT) algorithm for comparative validation. The same collocation criteria (in Section VI-A) as those used for POSP AOD were applied to the MODIS DT AOD. Table III shows the comparison results. It is evident that, within the same time period (for the whole year of 2022), the retrieval accuracy of POSP AOD is significantly higher than that of MODIS DT AOD. The RMSEs of POSP AOD and MODIS DT AOD are 0.061 and 0.088, respectively, and the fractions within the GCOS requirement are 66.7% and 49.2%, respectively. Overall, POSP AOD demonstrates significantly higher accuracy compared to MODIS DT AOD.

C. Effect of Adding Polarization for Aerosol Parameters Retrieval

To quantitatively assess the effect of adding polarization for aerosol retrieval, the retrievals from POSP/GF-5(02) data with and without polarization have been validated. Table IV presents the comparison of retrieval accuracy. For all aerosol optical properties, the retrieval accuracy significantly improved after adding the polarization. For AOD_{550}, the RMSE and MAE decreased from 0.121 and 0.084 to 0.061 and 0.040, respectively, and the fraction within the GCOS requirement increased from 33.7% to 66.7%. For $AE_{670-865}$, the RMSE and MAE decreased from 0.506 and 0.402 to 0.479 and 0.379, respectively. For SSA_{550}, the RMSE and MAE decreased from 0.045 and 0.037 to 0.037 and 0.028, respectively, and the fraction within the GCOS requirement increased from 49.6% to 60.3%.

Moreover, the bias decreased for all parameters (except SSA_{550}), making the retrieval results closer to being unbiased. Therefore, the addition of polarization constraints significantly reduces retrieval uncertainty.

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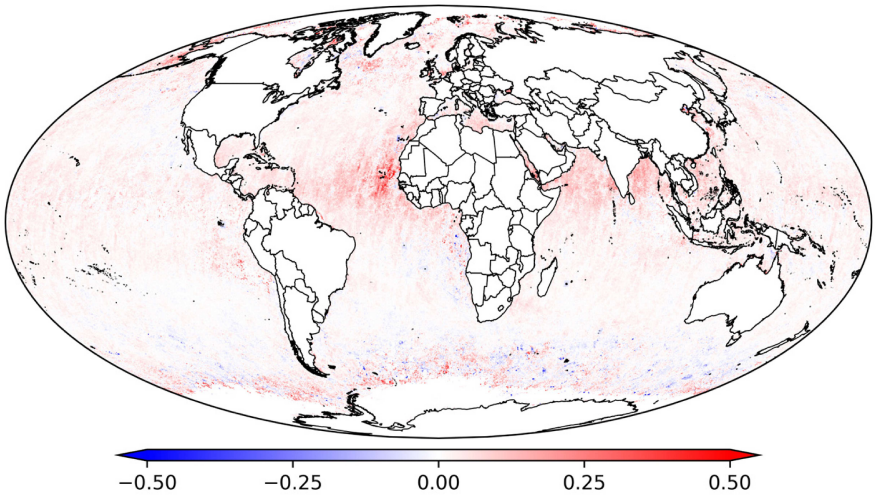


Fig. 12. Spatial distribution of the difference in AOD at 550 nm between POSP/GF-5(02) and MODIS/Terra from January 1, 2022 to December 31, 2022.

TABLE IV
COMPARISON OF STATISTICAL METRICS FOR RETRIEVED AEROSOL PARAMETERS FROM POSP/GF-5(02) DATA WITH AND WITHOUT POLARIZATION

Parameter	Retrieval utilized intensity and polarization				Retrieval utilized intensity only			
	RMSE	MAE	Bias	GCOS (%)	RMSE	MAE	Bias	GCOS (%)
AOD_{550}	0.061	0.040	0.010	66.7%	0.121	0.084	0.051	33.7%
$AE_{670\text{-}865}$	0.479	0.379	-0.004	-	0.506	0.402	-0.043	-
SSA_{550}	0.037	0.028	-0.004	60.3%	0.045	0.037	-0.003	49.6%

TABLE V
COMPARISON BETWEEN STATISTICAL METRICS OF AEROSOL PROPERTIES RETRIEVED FROM POSP USING THIS STUDY AND THE POSP/GRASP V1.0 PRODUCT

Parameter	This study			POSP/GRASP V1.0		
	RMSE	Bias	GCOS (%)	RMSE	Bias	GCOS (%)
AOD	0.061	0.010	66.7%	0.091	0.008	54.3%
AE	0.479	-0.004	-	0.667	-0.048	-

D. Comparison of Retrieval Accuracy Between This Study and GRASP

Chen et al. [71] successfully retrieved aerosol properties from POSP data for the year 2022 using the generalized retrieval of atmosphere and surface property (GRASP)/models algorithm. We also compared the performance of our algorithm with that of the GRASP algorithm over the same time period. It should be noted that the validation of POSP/GRASP results is obtained based on Level 2.0 AEROET data. This study presents a preliminary comparison, and we will conduct more detailed comparisons in future work. Furthermore, Chen et al. [71] did not provide a separate validation of SSA over the ocean; we compared only the statistical metrics of AOD and AE.

Table V presents the comparison results, which show that the aerosol optical properties retrieved using our algorithm exhibit comparable and even sometimes slightly higher accuracy than POSP/GRASP products. For AOD, our algorithm demonstrates a slightly lower RMSE and a higher fraction of retrievals meeting the GCOS requirements. For AE, our algorithm also achieves a lower RMSE.

Previous studies have shown that GRASP achieves high retrieval accuracy when applied to multiview polarimeters [66]; our results are comparable to those of GRASP. Therefore, the comparison results indicate that the algorithm proposed in this study effectively addresses the limitations of single-view observations and achieves comparable retrieval accuracy with the GRASP algorithm.

VIII. CONCLUSION

We present a high-accuracy aerosol retrieval algorithm tailored for spaceborne single-view polarimeters over the ocean. By replacing the traditional LUTs with an NN model, the algorithm achieves substantial gains in efficiency while preserving retrieval accuracy.

The validation of POSP and AERONET data demonstrates the capability to support accurate aerosol retrieval. The retrieval framework is theoretically portable and can be applied to other sensors with adjustments to RT calculations. Validation with AERONET and MAN shows that the retrieved AOD, AE, and SSA achieve high accuracy. Furthermore, for AOD_{550} and SSA_{550}, 66.7% and 60.3% of retrievals meet GCOS requirements. In addition, the validation with MAN exhibits slightly better performance than the validation with AERONET. The results confirm that incorporating polarization information yields markedly higher accuracy than retrievals with intensity-only and outperform MODIS AOD retrievals. Furthermore, comparison with the GRASP algorithm indicates

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that our algorithm delivers comparable, and in some cases superior, accuracy.

These findings highlight the essential role of polarization in advancing aerosol remote sensing and demonstrate the feasibility of developing specialized retrieval strategies for single-view polarimeters, thereby paving the way for next-generation satellite aerosol observations.

ACKNOWLEDGMENT

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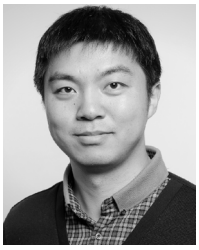
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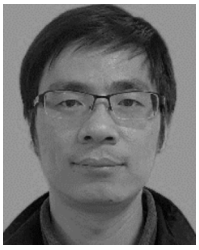
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